

ENCROACHING ON THE MARGIN

Urban Sprawl and Compound Flood Hazards in Lagos

Along the Gulf of Guinea, Lagos State stands as a stark visual testament to the precarious balance between rapid urbanization and ecological resilience. Spanning $3,577\text{km}^2$, this coastal megacity faces an existential threat from climate-induced sea-level rise, compounding pluvial flooding, and unchecked urban sprawl. By synthesizing multi-temporal Landsat Surface Reflectance imagery, JRC Surface Water baselines, and NASA DEM topography in Google Earth Engine and QGIS, this Earth Observation product maps the systematic degradation of natural coastal buffers from 2000 to 2025.

The spatial evidence is striking. Low-lying terrain under 5 meters in elevation covers 49% ($1,671\text{km}^2$) of the state, defining an inherently vulnerable, waterlogged landscape. Driven by demographic growth, built-up urban footprints expanded to cover 27% (930km^2) of the land area, progressively encroaching upon 784km^2 of historical wetland buffers. Crucially, the spatial intersection analysis uncovers 5.6km^2 of acute "collision hotspots" critical zones where high-density human settlements directly overwrite natural drainage channels and coastal swamps.

This map captures a brooding environmental crisis: as human development paving over low-lying wetlands accelerates, it destroys the ecosystem services that historically absorbed surge waters. In their place, concrete infrastructure now traps runoff, amplifying flood exposure across vulnerable communities.

Ultimately, this single image serves as both a spatial diagnosis and an urgent call to action. It demonstrates how Earth Observation data can illuminate compound climate risks, urging coastal megacities worldwide to prioritize nature-based solutions, wetland restoration, and climate-resilient spatial planning before the water reclaims its margin.